SECTION 5

Experimental Results

The paper's plots, discussed — and where they open the door to our extension

Quadruple-tank (§5.2) · Furuta pendulum (§5.3) · Figures 2–4 · Table 1

Optimisation

A Mixed-Integer Quadratically-Constrained Program (MIQCP) in Gurobi (`gurobipy`), with indicator constraints for the hit/miss dynamics.

What is measured

Per configuration: the optimal attack schedule, the optimal and nominal objectives, and effectiveness $\omega = J_{\text{attack}} - J_{\text{nominal}}$.

Swept parameters

- attack horizon $\ell \in \{1, \dots, 25\}$
- watchdog threshold $\bar{a} \in \{1, \dots, 5\}$
- detection window $n_w \in \{10, \dots, 20\}$
- nominal miss prob. $\hat{p}_{\text{m}} \in \{0.01, \dots, 0.25\}$
- detector: binomial · χ^2 · combined

Note: the full sweep is expensive (hours–days with a Gurobi licence).

Quadruple-tank process

§5.2 System A — stable, non-minimum-phase, simulated (HOLD)

Setup

- Four interconnected tanks; two pumps, cross-coupled.
- Non-minimum-phase configuration (hard to control).
- LQG controller; HOLD policy on a miss; $T_s = 0.1$ s.
- Open-loop **stable** — evaluated in simulation.
- Objective weights the two lower-tank outputs.

What the attack does here

Because the plant is stable, delay does not destabilise it — it **degrades regulation**: levels drift from setpoint and settle slowly, bounded and recoverable. Damage is real but far smaller than on the pendulum.

Furuta rotary pendulum

§5.3 System B — unstable, LQR, HOLD & ZERO, real hardware

Setup

- Rotary arm + pendulum; controller balances it upright.
- Open-loop **unstable** — the high-stakes case.
- LQR controller; **both** HOLD and ZERO tested; $T_s = 5\,\text{ms}$.
- Validated on **real hardware** (with a video).
- Objective weights the pendulum angle.

What the attack does here

A well-timed stealthy burst of misses under HOLD drives the pendulum off upright and can **topple it**. ZERO, the mitigation, keeps it near upright under the same attack.

Figure 2 — effectiveness vs attack length

Both systems · $\bar{a} = 1$–4 · combined binomial + χ^2 detector, $n_w = 20$

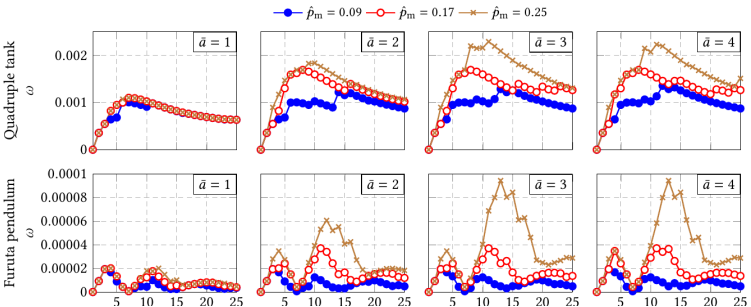


Figure 2 · © ACM — Nauta, Sandberg & Maggio, ICCPS 2025

Discuss: x-axis is attack length ℓ (1–25); columns are $\bar{a} = 1$–4; top row the quadruple tank (HOLD), bottom the Furuta (ZERO). ω peaks around $\ell \approx 10$ — longer attacks are not always worse — and raising $\bar{a}$ beyond 3 barely changes the impact. → **Extension hook:** *the watchdog saturates by $\bar{a} \approx 3$, so a realistic threshold cannot stop this — we optimise the DEFENCE configuration instead.*

Figure 3 — detector comparison

binomial vs combined vs geometric (χ^2) · $\bar{a} = 4$, $n_w = 20$

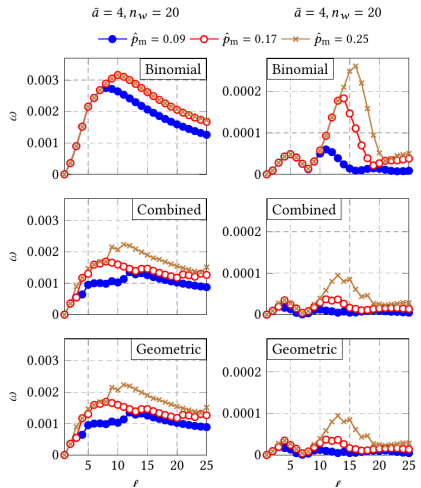


Figure 3 · © ACM — Nauta, Sandberg & Maggio, ICCPS

Read it

- Rows: binomial, combined, geometric detector.
- Left column = quadruple tank (HOLD); right = Furuta (ZERO).
- x-axis = ℓ; curves for $\hat{p}_m = 0.09/0.17/0.25$.
- Geometric (χ^2) and combined limit the attacker **more** than binomial alone.
- The long-term χ^2 test catches rare long bursts the short window misses.

→ **Extension hook:** *χ^2 is the strongest single detector — our extension stacks multiple simultaneous windows + combined detection to push ω lower still.*

Figure 4 — ZERO vs HOLD (Furuta)

varying window $n_w = 10$–18 · $\bar{a} = 4$ · binomial detector

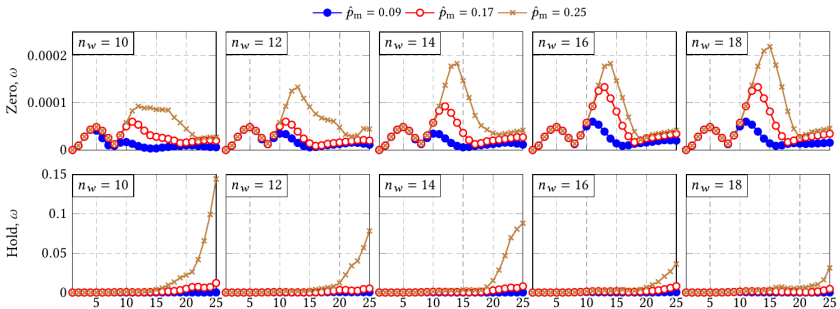


Figure 4 · © ACM — Nauta, Sandberg & Maggio, ICCPS 2025

Discuss: Top row ZERO ($\omega \approx 2 \times 10^{-4}$); bottom row HOLD ($\omega$ up to 0.15 — note the far larger scale). Under HOLD a well-timed late burst **destabilises** the pendulum; ZERO keeps ω tiny. Shorter windows help HOLD; longer windows permit longer bursts. → **Extension hook:** *the safe policy is plant-dependent — our defender sweep picks the policy + threshold that minimise worst-case ω.*

Table 1 — optimised attack sequences

Furuta HOLD · $\hat{p}_{\mathrm{m}} = 0.25$, $\bar{a} = 4$, $\ell = 25$ · *varying window* n_w

n_w	Attack	% in $\mathcal{W}$	$\sum_i (a_i)$
10	1110111000111000111011100	60%	15
12	1011110001101000111101100	58%	14
14	1101111000110001111011000	57%	14

Table 1 · © ACM — Nauta, Sandberg & Maggio, ICCPS 2025

Discuss: each row gives the optimal attack schedule, the % of deadline misses allowed in one window (% in $\mathcal{W}$), and the total misses $\sum_i a_i$. As the window widens (10→14), the allowed miss density drops (60%→57%) — a wider window forces a sparser attack (the binomial test allows 7 misses in a window of 12, 8 in a window of 14). → **Extension hook:** *every sequence here optimises the ATTACKER against a FIXED defence — our novelty flips this and optimises the defence itself.*

Section 5 — key findings

Four results from the experiments

(i) Impact isn't monotonic in ℓ

ω peaks around $\ell \approx 10$; the optimiser concentrates or interleaves delays by initial conditions and budget.

(ii) Watchdogs barely help

Effective only at $\bar{a} = 1$–2; realistic industrial thresholds (3–10) give little protection.

(iii) χ^2 dominates the binomial test

The long-term run-length test limits the attacker far more than the sliding window.

(iv) ZERO beats HOLD (unstable)

For the Furuta pendulum, zeroing the input on a miss is a far better defence; window length trades off per policy.

FROM THE PAPER'S RESULTS → OUR EXTENSION

What the figures leave open — and what we add

What §5 establishes

- Watchdog is weak at realistic $\bar{a}$ (Fig 2)
- χ^2 is the strongest single detector (Fig 3)
- ZERO beats HOLD on the unstable plant (Fig 4)
- All optimised for a **fixed** defence (Table 1)

Our extension (defender side)

- Optimise the **defence**, not the attacker
- Sweep (policy, $\bar{a}$, window, $\hat{p}_{\mathrm{m}}$, α) $\rightarrow \omega_{\mathrm{worst}}$
- Rank configs that cap worst-case damage
- Multi-window detector + live visual demo

The gap we fill: the paper asks "how bad can a stealthy attacker be against a given defence?" We answer the defender's question — "which defence minimises the worst stealthy attacker?" — and show it live.